

RatedCare Connect (RCC)

Healthcare Workforce Management & Nurse Rostering SaaS

Role	Full Stack MERN Developer
Project Type	Healthcare SaaS Platform
Industry	Healthcare / Workforce Management
Technology	React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, Anthropic Claude AI, REST APIs, DigitalOcean, Nginx, PM2, GitHub Actions

Project Overview

RatedCare Connect (RCC) is a comprehensive healthcare workforce management platform designed to streamline the operational management of nurses, practitioners, staff, rosters, attendance, leave, payroll, workforce capacity, and revenue.

The platform brings multiple healthcare workforce operations into a centralized system, allowing administrators and healthcare organizations to manage staff, create and monitor duty rosters, track attendance, manage leave balances, monitor compliance documents, calculate workforce-related metrics, and analyze operational performance through dashboards.

The system was developed as a production SaaS application with a focus on **scalability, secure role-based access, multi-tenant architecture, automation, real-time operational workflows, and production deployment**.

The Challenge

Healthcare organizations often manage workforce operations across disconnected processes. Roster planning, nurse availability, attendance, leave balances, payroll calculations, compliance documents, and workforce analytics can become difficult to manage when handled manually or through separate systems. RCC was designed to bring these workflows together into one centralized platform.

The core objective was to create a system that could help healthcare organizations:

- Manage businesses and healthcare organizations
- Register and manage users, nurses, practitioners, and staff
- Create and manage duty rosters
- Track employee attendance
- Support GPS-based check-in and check-out
- Monitor duty hours
- Manage leave balances and leave-related workflows

- Manage employee visa information and expiry tracking
- Support payroll-related operations
- Monitor FTE and workforce capacity
- Track revenue-related information
- Provide operational dashboards and analytics
- Automate document verification using AI
- Maintain secure tenant-level data isolation

My Role

I worked across the full application lifecycle as a **Full Stack MERN Developer**, contributing to both frontend and backend development as well as production infrastructure.

My responsibilities included:

- Designing and developing frontend interfaces with React.js
- Developing RESTful APIs using Node.js and Express.js
- Designing MongoDB schemas and data models
- Implementing authentication and authorization
- Developing role-based access control
- Implementing multi-tenant data isolation
- Building workforce and roster management workflows
- Developing attendance and GPS check-in/out functionality
- Developing leave balance management
- Building visa management and expiry tracking
- Implementing AI-powered document verification
- Developing dashboard analytics
- Integrating third-party services
- Testing APIs using Postman
- Managing production deployments
- Configuring Linux, Nginx, and PM2
- Managing GitHub Actions and CI/CD workflows
- Troubleshooting production, server, database, and API issues

Core Platform Modules

1. Workforce & User Management

RCC provides centralized workforce management for healthcare organizations. The system supports management of different workforce entities including businesses/healthcare organizations, users, staff, nurses, practitioners, and workforce assignments. The platform provides controlled access based on user roles and

organizational ownership, allowing different users to interact only with the areas relevant to their responsibilities.

2. Multi-Tenant Architecture

One of the important architectural requirements of RCC was maintaining strict separation between healthcare organizations. The platform follows a secure multi-tenant approach where HR and organizational users are restricted to the data belonging to their respective organization, preventing access to workforce records, rosters, leave information, or other operational data belonging to another tenant.

- Tenant-level data isolation
- Role-based authorization
- Secure API access
- Organization-specific workforce management
- Protected administrative operations

3. Duty Roster Management

Roster management is one of the core features of RCC. Administrators and authorized users can organize workforce schedules and assign duties according to operational requirements.

- Creating duties
- Assigning staff and nurses
- Managing duty schedules
- Viewing workforce assignments
- Monitoring scheduled hours
- Managing roster information
- Supporting monthly duty views
- Connecting roster data with attendance workflows

The system provides healthcare staff with a centralized view of their scheduled duties instead of relying on manually maintained schedules.

4. Nurse Mobile Workflow

RCC also supports workforce operations through a mobile-oriented experience for nurses. Nurses can access their scheduled duties and review their monthly roster, staying informed about assigned duties, duty dates, scheduled working periods, and check-in/out requirements. This connects the workforce scheduling process with real-world attendance operations.

5. GPS Check-In & Check-Out

RCC includes location-aware attendance functionality. Healthcare staff can check in and check out from their assigned duties using the platform, capturing attendance against scheduled workforce activity while incorporating GPS/location information.

- Staff arrival

- Staff departure
- Attendance records
- Duty hours
- Scheduled vs actual workforce activity

6. Attendance Management

The attendance system provides administrators with visibility into employee attendance and working activity, connecting roster assignments with actual attendance records.

- Check-in records
- Check-out records
- Duty hours
- Attendance status
- Workforce attendance monitoring
- Administrative attendance review

7. Payroll & Duty Hours

RCC includes payroll-related workforce management capabilities connected with staff duties and working hours, allowing administrators to review workforce activity and duty hours that contribute to payroll operations.

- Employee duty hours
- Working time
- Payroll-related workforce information
- Staff salary information
- Workforce cost analysis

8. Leave Management

The HR Leave Module was developed to support leave management for both Staff and Users, while maintaining compatibility with the existing RCC Admin Dashboard.

- Individual leave allocation
- Bulk leave allocation
- Leave balance updates
- Manual balance adjustments
- Automatic leave balance calculations
- Leave deduction
- Leave restoration
- Year-based leave balances

The architecture was also designed to maintain tenant-level security so HR users can manage leave information only within their authorized healthcare organization.

9. Visa Management & Compliance Tracking

RCC also includes a Visa Management module for workforce compliance tracking.

- Visa record management
- Employee/user mapping
- Secure document uploads
- Visa expiry tracking
- Visa status management
- Dashboard analytics
- Upcoming expiry monitoring
- API-based management workflows

10. AI-Powered Document Verification

One of the advanced capabilities of RCC is AI-powered document verification. Anthropic Claude was integrated into the platform to assist with document verification workflows. The AI-powered pipeline processes and analyzes workforce documentation, reducing the amount of manual verification required from administrators.

- Document processing
- AI-based verification
- Backend API integration
- Validation workflows
- Verification results
- Integration with workforce records

11. FTE Management

RCC includes workforce capacity management around Full-Time Equivalent (FTE) information, providing organizations with better visibility into workforce capacity and staffing requirements.

- Workforce allocation
- Staff capacity
- Working hours
- Staffing requirements
- Workforce utilization

12. Revenue Management & Analytics

Revenue-related information is integrated into the RCC administrative environment. The dashboard provides visibility into operational and financial metrics, connecting:

Workforce → Duties → Attendance → Hours → FTE → Revenue

13. Admin Dashboard

The RCC Admin Dashboard acts as the central control center for the platform, giving administrators visibility into important operational information from a centralized interface.

- Workforce information
- Duty rosters
- Attendance
- Check-in/check-out activity
- Duty hours
- Payroll-related information
- FTE
- Revenue
- Messages
- Workforce activity
- Upcoming document expiries

14. Messaging & Notifications

The platform provides messaging and notification-related functionality for workforce operations, connected to workflows such as:

- Workforce activities
- Roster information
- Administrative updates
- Document expiry monitoring
- Operational communication

15. Role-Based Access Control

Security was a major consideration throughout the platform. RCC uses authentication and role-based authorization to control access to different parts of the system, ensuring administrative functionality is protected, HR functionality is restricted, workforce users access only relevant information, tenant data remains isolated, and protected API endpoints cannot be accessed by unauthorized users. JWT-based authentication was used as part of the API security architecture.

16. Backend Architecture

The backend was developed using a modular Node.js and Express.js architecture, following structured separation between models, repositories, services, controllers, validation, routes, and middleware. MongoDB and Mongoose were used for database management and data modeling, with a RESTful API layer for frontend and workforce workflows.

17. API Development & Testing

I developed and tested APIs for modules including:

- Workforce management
- Staff
- Users
- Rosters
- Attendance
- Leave balances
- Visa management
- Dashboard analytics
- AI verification
- Payroll-related workflows

Postman was used extensively for API testing and validation, verifying:

- Request validation
- Authentication
- Authorization
- Successful responses
- Error handling
- Database operations
- Role restrictions
- Tenant restrictions

18. Production Infrastructure

RCC was deployed and maintained in a production environment rather than remaining as a local development project, using DigitalOcean, Ubuntu/Linux, Nginx, PM2, MongoDB Atlas, Git, GitHub Actions, and CI/CD workflows.

Responsibilities included resolving deployment issues, server configuration problems, permission issues, API errors, database issues, application crashes, and process management issues.

19. CI/CD & Deployment

GitHub Actions was used to support automated deployment workflows, connecting source control with the production environment and reducing repetitive manual deployment operations.

Development → Git → GitHub → CI/CD → Server → Nginx → PM2 → Production

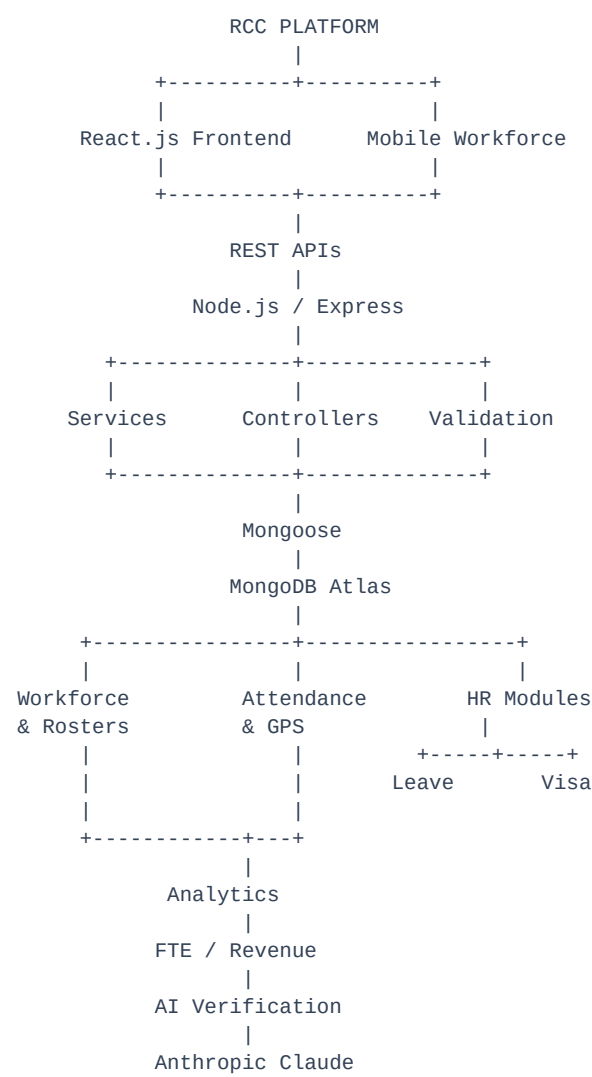
Technology Stack

Frontend

React.js, JavaScript, Tailwind CSS, Responsive UI, Dashboard interfaces

Backend	Node.js, Express.js, REST APIs, JWT Authentication, Middleware, Service-based architecture
Database	MongoDB, MongoDB Atlas, Mongoose, Data modeling
AI	Anthropic Claude, AI Document Verification, AI-powered processing workflows
DevOps / Infra	DigitalOcean, Ubuntu/Linux, Nginx, PM2, GitHub Actions, CI/CD
Dev & Testing	Git, GitHub, npm, Postman

Architecture Overview



Key Engineering Challenges

Multi-Tenant Security

The platform required strict separation between organizations. This meant API-level authorization and tenant-aware database operations had to be consistently applied across modules.

Complex Workforce Relationships

RCC connects users, nurses, staff, duties, rosters, attendance, leave, payroll, FTE, and revenue. Designing these relationships required careful data modeling and backend logic.

Connecting Planned vs Actual Workforce Activity

Roster data represents planned work while attendance represents actual activity. Connecting these workflows provides a more accurate view of workforce utilization and duty hours.

AI Integration

Introducing AI document verification into the platform required integrating an external AI service with existing backend workflows while maintaining validation and structured application data.

Production Reliability

RCC operates as a production system, requiring ongoing attention to deployment, server configuration, process management, database operations, permissions, API failures, and application debugging.

Key Outcomes

RCC transformed multiple healthcare workforce processes into a centralized digital platform, providing organizations with:

- Centralized workforce management
- Digital nurse rostering
- Monthly duty visibility
- GPS-based attendance
- Check-in/check-out tracking
- Duty-hour monitoring
- Payroll-related workforce data
- Leave balance management
- Visa and document expiry tracking
- AI-powered document verification
- FTE management
- Revenue visibility
- Role-based access control
- Multi-tenant security
- Operational dashboards

- Production cloud infrastructure

My Contribution

My work on RCC covered significantly more than frontend development. I contributed across the complete software lifecycle:

Frontend Dev → Backend Dev → Database Design → API Dev → Authentication → Authorization → AI Integration → Module Dev → API Testing → Deployment → CI/CD → Production Support

This project gave me hands-on experience building and maintaining a real-world healthcare SaaS platform where **security, scalability, workforce operations, AI integration, and production reliability** were all critical requirements.

Project Impact

RatedCare Connect demonstrates how modern web technologies can be used to centralize complex healthcare workforce operations into a single scalable platform.

Instead of treating rostering, attendance, HR, compliance, payroll, workforce capacity, and analytics as isolated processes, RCC connects them into a unified operational ecosystem.

The result is a production-focused SaaS platform capable of supporting the day-to-day workforce management needs of healthcare organizations while providing a foundation for further automation and AI-powered workflows.